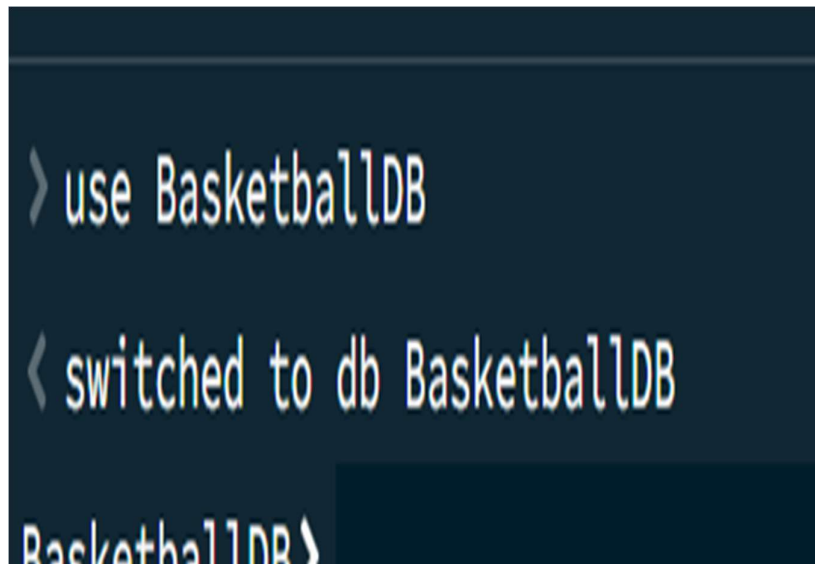


## 1. CREATE DATABASE

MongoDB Query:

use BasketballDB

**Explanation:** Creates or switches to the BasketballDB database.



```
> use BasketballDB
< switched to db BasketballDB
BasketballDB>
```

## 2. CREATE COLLECTION

MongoDB Query:

```
db.createCollection("matches")
```

**Explanation:** Creates a collection named matches.

```
> db.createCollection("matches")  
< { ok: 1 }  
BasketballDB>
```

### 3. INSERT 50 BASKETBALL MATCH RECORDS

#### MongoDB Query:

```
db.matches.insertMany([
  {matchId:1, team1:"India", team2:"Australia", team1Score:92, team2Score:87, venue:"New Delhi"},
  {matchId:2, team1:"USA", team2:"Canada", team1Score:105, team2Score:98, venue:"Toronto"},
  {matchId:3, team1:"Spain", team2:"France", team1Score:88, team2Score:91, venue:"Madrid"},
  {matchId:4, team1:"Germany", team2:"Italy", team1Score:96, team2Score:82, venue:"Berlin"},
  {matchId:5, team1:"India", team2:"Japan", team1Score:101, team2Score:95, venue:"Mumbai"},
  {matchId:6, team1:"Brazil", team2:"Argentina", team1Score:89, team2Score:94, venue:"Rio de Janeiro"},
  {matchId:7, team1:"Australia", team2:"China", team1Score:97, team2Score:84, venue:"Sydney"},
  {matchId:8, team1:"France", team2:"Belgium", team1Score:91, team2Score:79, venue:"Paris"},
  {matchId:9, team1:"USA", team2:"Mexico", team1Score:112, team2Score:89, venue:"Dallas"},
  {matchId:10, team1:"Canada", team2:"Brazil", team1Score:86, team2Score:90, venue:"Vancouver"},

  {matchId:11, team1:"India", team2:"China", team1Score:98, team2Score:93, venue:"Kolkata"},
  {matchId:12, team1:"Japan", team2:"South Korea", team1Score:87, team2Score:82, venue:"Tokyo"},
  {matchId:13, team1:"Spain", team2:"Germany", team1Score:94, team2Score:89, venue:"Barcelona"},
  {matchId:14, team1:"Italy", team2:"Greece", team1Score:81, team2Score:88, venue:"Rome"},
  {matchId:15, team1:"Australia", team2:"Japan", team1Score:103, team2Score:91, venue:"Melbourne"},
  {matchId:16, team1:"Argentina", team2:"Chile", team1Score:95, team2Score:78, venue:"Buenos Aires"},
  {matchId:17, team1:"USA", team2:"France", team1Score:108, team2Score:102, venue:"New York"},
  {matchId:18, team1:"Canada", team2:"Spain", team1Score:90, team2Score:97, venue:"Montreal"},
  {matchId:19, team1:"India", team2:"South Korea", team1Score:85, team2Score:80, venue:"Bengaluru"}])
```

{matchId:20, team1:"Brazil", team2:"Uruguay", team1Score:92, team2Score:76, venue:"Sao Paulo"},

{matchId:21, team1:"Germany", team2:"France", team1Score:99, team2Score:101, venue:"Munich"},

{matchId:22, team1:"Australia", team2:"India", team1Score:94, team2Score:89, venue:"Sydney"},

{matchId:23, team1:"USA", team2:"Spain", team1Score:115, team2Score:108, venue:"Los Angeles"},

{matchId:24, team1:"Japan", team2:"China", team1Score:83, team2Score:96, venue:"Osaka"},

{matchId:25, team1:"Argentina", team2:"Brazil", team1Score:91, team2Score:87, venue:"Cordoba"},

{matchId:26, team1:"France", team2:"Italy", team1Score:104, team2Score:92, venue:"Lyon"},

{matchId:27, team1:"Canada", team2:"USA", team1Score:88, team2Score:109, venue:"Toronto"},

{matchId:28, team1:"India", team2:"Nepal", team1Score:110, team2Score:72, venue:"New Delhi"},

{matchId:29, team1:"Spain", team2:"Portugal", team1Score:93, team2Score:85, venue:"Madrid"},

{matchId:30, team1:"Germany", team2:"Netherlands", team1Score:87, team2Score:90, venue:"Hamburg"},

{matchId:31, team1:"Australia", team2:"New Zealand", team1Score:102, team2Score:96, venue:"Perth"},

{matchId:32, team1:"India", team2:"Sri Lanka", team1Score:99, team2Score:76, venue:"Chennai"},

{matchId:33, team1:"USA", team2:"Argentina", team1Score:118, team2Score:97, venue:"Chicago"},

{matchId:34, team1:"France", team2:"Spain", team1Score:86, team2Score:93, venue:"Paris"},

{matchId:35, team1:"Brazil", team2:"Colombia", team1Score:100, team2Score:84, venue:"Brasilia"},

{matchId:36, team1:"Japan", team2:"Philippines", team1Score:89, team2Score:94, venue:"Tokyo"},

{matchId:37, team1:"Germany", team2:"Belgium", team1Score:105, team2Score:88, venue:"Frankfurt"},

{matchId:38, team1:"Canada", team2:"Mexico", team1Score:96, team2Score:81, venue:"Vancouver"},

{matchId:39, team1:"India", team2:"Iran", team1Score:91, team2Score:86, venue:"Hyderabad"},

{matchId:40, team1:"Australia", team2:"South Korea", team1Score:108, team2Score:92, venue:"Melbourne"},

{matchId:41, team1:"USA", team2:"Germany", team1Score:111, team2Score:95, venue:"Boston"},

{matchId:42, team1:"Spain", team2:"Greece", team1Score:97, team2Score:89, venue:"Valencia"},

```
{matchId:43, team1:"India", team2:"Japan", team1Score:102, team2Score:88,  
venue:"Pune"},  
{matchId:44, team1:"Brazil", team2:"Chile", team1Score:93, team2Score:79, venue:"Rio de  
Janeiro"},  
{matchId:45, team1:"France", team2:"Germany", team1Score:100, team2Score:98,  
venue:"Marseille"},  
{matchId:46, team1:"Australia", team2:"Philippines", team1Score:107, team2Score:99,  
venue:"Brisbane"},  
{matchId:47, team1:"Canada", team2:"Argentina", team1Score:89, team2Score:92,  
venue:"Ottawa"},  
{matchId:48, team1:"India", team2:"Australia", team1Score:106, team2Score:101,  
venue:"Ahmedabad"},  
{matchId:49, team1:"USA", team2:"Canada", team1Score:120, team2Score:104, venue:"Las  
Vegas"},  
{matchId:50, team1:"Spain", team2:"Italy", team1Score:98, team2Score:83, venue:"Seville"}  
])
```

**Explanation:** Inserts 50 basketball match documents into the matches collection

MONGOSH

```
'27': ObjectId('6ac69002a44099e363056789'),
'28': ObjectId('6ac69002a44099e36305678a'),
'29': ObjectId('6ac69002a44099e36305678b'),
'30': ObjectId('6ac69002a44099e36305678c'),
'31': ObjectId('6ac69002a44099e36305678d'),
'32': ObjectId('6ac69002a44099e36305678e'),
'33': ObjectId('6ac69002a44099e36305678f'),
'34': ObjectId('6ac69002a44099e363056790'),
'35': ObjectId('6ac69002a44099e363056791'),
'36': ObjectId('6ac69002a44099e363056792'),
'37': ObjectId('6ac69002a44099e363056793'),
'38': ObjectId('6ac69002a44099e363056794'),
'39': ObjectId('6ac69002a44099e363056795'),
'40': ObjectId('6ac69002a44099e363056796'),
'41': ObjectId('6ac69002a44099e363056797'),
'42': ObjectId('6ac69002a44099e363056798'),
'43': ObjectId('6ac69002a44099e363056799'),
'44': ObjectId('6ac69002a44099e36305679a'),
'45': ObjectId('6ac69002a44099e36305679b'),
'46': ObjectId('6ac69002a44099e36305679c'),
'47': ObjectId('6ac69002a44099e36305679d'),
'48': ObjectId('6ac69002a44099e36305679e'),
'49': ObjectId('6ac69002a44099e36305679f')
```

## 4. DISPLAY ALL PLAYERS

**MongoDB Query:**

```
db.matches.find()
```

**Explanation:** Retrieves all documents from the matches collection.

```
> db.matches.find()
< {
  _id: ObjectId('6ac69002a44099e36305676e'),
  matchId: 1,
  team1: 'India',
  team2: 'Australia',
  team1Score: 92,
  team2Score: 87,
  venue: 'New Delhi'
}
{
  _id: ObjectId('6ac69002a44099e36305676f'),
  matchId: 2,
  team1: 'USA',
  team2: 'Canada',
  team1Score: 105,
  team2Score: 98,
  venue: 'Toronto'
}
```

## 5. EQUAL TO — \$eq

MongoDB Query:

```
db.matches.find({
  team1Score: {$eq: 92}
```

})

**Explanation:** Finds basketball matches where Team 1 scored exactly 92 points.

```
type: 'C' for more
> db.matches.find({
  team1Score: {$eq: 92}
})
< {
  _id: ObjectId('6ac69002a44099e36...'),
  matchId: 1,
  team1: 'India',
  team2: 'Australia',
  team1Score: 92,
  team2Score: 87,
  venue: 'New Delhi'
}
{
  _id: ObjectId('6ac69002a44099e36...'),
  matchId: 20,
  team1: 'Brazil',
  team2: 'Uruguay',
  team1Score: 92,
```

## 6. GREATER THAN — \$gt

MongoDB Query:

```
db.matches.find({
  team1Score: {$gt: 100}
```

```
})
```

**Explanation:** Finds basketball matches where Team 1 scored more than 100 points.

```
> db.matches.find({
  team1Score: {$gt: 100}
})
< {
  _id: ObjectId('6ac69002a44099e36305676f'),
  matchId: 2,
  team1: 'USA',
  team2: 'Canada',
  team1Score: 105,
  team2Score: 98,
  venue: 'Toronto'
}
{
  _id: ObjectId('6ac69002a44099e363056772'),
  matchId: 5,
  team1: 'India',
  team2: 'Japan',
  team1Score: 101,
  team2Score: 95,
  venue: 'Mumbai'
}
```

## 7. LESS THAN — \$lt

MongoDB Query:

```
db.matches.find({
  team1Score: {$lt: 90}
})
```

**Explanation:** Finds basketball matches where Team 1 scored less than 90 points.

```
> db.matches.find({
  team1Score: {$lt: 90}
})
< {
  _id: ObjectId('6ac69002a44099e363056770'),
  matchId: 3,
  team1: 'Spain',
  team2: 'France',
  team1Score: 88,
  team2Score: 91,
  venue: 'Madrid'
}
{
  _id: ObjectId('6ac69002a44099e363056773'),
  matchId: 6,
  team1: 'Brazil',
  team2: 'Argentina',
  team1Score: 89,
  team2Score: 94,
  venue: 'Rio de Janeiro'
}
{
```

## 8. GREATER THAN OR EQUAL TO — \$gte

MongoDB Query:

```
db.matches.find({
  team1Score: {$gte: 100}
```

```
})
```

**Explanation:** Finds basketball matches where Team 1 scored 100 or more points.

```
    venue: 'Ottawa'
  }
> db.matches.find({
  team1Score: {$gte: 100}
})
< {
  _id: ObjectId('6ac69002a44099e36305676f'),
  matchId: 2,
  team1: 'USA',
  team2: 'Canada',
  team1Score: 105,
  team2Score: 98,
  venue: 'Toronto'
}
{
  _id: ObjectId('6ac69002a44099e363056772'),
  matchId: 5,
  team1: 'India',
  team2: 'Japan',
  team1Score: 101,
  team2Score: 95,
  venue: 'Mumbai'
}
{
```

## 9. LESS THAN OR EQUAL TO — \$lte

MongoDB Query:

```
db.matches.find({
  team1Score: {$lte: 90}
})
```

**Explanation:** Finds basketball matches where Team 1 scored 90 or fewer points.

```
}
> db.matches.find({
  team1Score: {$lte: 90}
})
< {
  _id: ObjectId('6ac69002a44099e363056770'),
  matchId: 3,
  team1: 'Spain',
  team2: 'France',
  team1Score: 88,
  team2Score: 91,
  venue: 'Madrid'
}
{
  _id: ObjectId('6ac69002a44099e363056773'),
  matchId: 6,
  team1: 'Brazil',
  team2: 'Argentina',
  team1Score: 89,
  team2Score: 94,
  venue: 'Rio de Janeiro'
}
{
```

## 10. UPDATE OPERATION

MongoDB Query:

```
db.matches.updateOne(
  {matchId:5},
  {$set:{team1Score:105}}
```

)

Explanation: Updates the Team 1 score to 105 for the match with matchId 5.

```
}
> db.matches.updateOne(
  {matchId:5},
  {$set:{team1Score:105}}
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

## 11. DELETE ONE OPERATION

**MongoDB Query:**

```
db.matches.deleteOne({
  matchId:10
})
```

**Explanation:** Deletes one document matching matchId 10.

```
}  
> db.matches.deleteOne({  
  matchId:10  
})  
< {  
  acknowledged: true,  
  deletedCount: 1  
}  
BasketballDB>
```

## 12. DELETE MANY OPERATION

**MongoDB Query:**

```
db.matches.deleteMany({  
  team1Score: {$lt:90}
```

```
})
```

**Explanation:** Deletes all basketball matches where Team 1 scored below 90 points.

```
}  
> db.matches.deleteMany({  
  team1Score: {$lt:90}  
})  
< {  
  acknowledged: true,  
  deletedCount: 11  
}  
BasketballDB>
```

## 13. AND OPERATION — \$and

**MongoDB Query:**

```
db.matches.find({  
  $and:[
```

```

    {team1Score:{$gt:100}},
    {team1:"India"}
  ]
})

```

**Explanation:** Finds matches satisfying both conditions: Team 1 is India and Team 1 scored more than 100 points.

```

}
db.matches.find({
  $and:[
    {team1Score:{$gt:100}},
    {team1:"India"}
  ]
})
{
  _id: ObjectId('6ac69002a44099e363056772'),
  matchId: 5,
  team1: 'India',
  team2: 'Japan',
  team1Score: 105,
  team2Score: 95,
  venue: 'Mumbai'
}
{
  _id: ObjectId('6ac69002a44099e363056789'),
  matchId: 28,
  team1: 'India',
  team2: 'Nepal',
  team1Score: 110,
  team2Score: 72,
  venue: 'New Delhi'
}
{

```

## 14. OR OPERATION — \$or

MongoDB Query:

```

db.matches.find({
  $or:[

```

```
{team1:"India"},  
{team1:"USA"}  
]  
})
```

**Explanation:** Finds basketball matches where either India or USA is Team 1.

```
> db.matches.find({  
  $or:[  
    {team1:"India"},  
    {team1:"USA"}  
  ]  
})  
< {  
  _id: ObjectId('6ac69002a44099e36305676e'),  
  matchId: 1,  
  team1: 'India',  
  team2: 'Australia',  
  team1Score: 92,  
  team2Score: 87,  
  venue: 'New Delhi'  
}  
{  
  _id: ObjectId('6ac69002a44099e36305676f'),  
  matchId: 2,  
  team1: 'USA',  
  team2: 'Canada',  
  team1Score: 105,  
  team2Score: 98,  
  venue: 'Toronto'  
}
```

## 15. SORT OPERATION

### A. Ascending Order

**MongoDB Query:**

```
db.matches.find().sort({  
  team1Score: 1  
})
```

**Explanation:** Sorts matches by Team 1's score from lowest to highest. Use 1 for ascending order.

```
}
> db.matches.find().sort({
  team1Score: 1
})
< {
  _id: ObjectId('6ac69002a44099e36305677f'),
  matchId: 18,
  team1: 'Canada',
  team2: 'Spain',
  team1Score: 90,
  team2Score: 97,
  venue: 'Montreal'
}
{
  _id: ObjectId('6ac69002a44099e363056775'),
  matchId: 8,
  team1: 'France',
  team2: 'Belgium',
  team1Score: 91,
  team2Score: 79,
  venue: 'Paris'
}
{
```

## B. Descending Order

**MongoDB Query:**

```
db.matches.find().sort({
  team1Score: -1
})
```

**Explanation:** Sorts matches by Team 1's score from highest to lowest. Use -1 for descending order.

```

Type "it" for more
> db.matches.find().sort({
  team1Score: -1
})
< {
  _id: ObjectId('6ac69002a44099e36305679e'),
  matchId: 49,
  team1: 'USA',
  team2: 'Canada',
  team1Score: 120,
  team2Score: 104,
  venue: 'Las Vegas'
}
{
  _id: ObjectId('6ac69002a44099e36305678e'),
  matchId: 33,
  team1: 'USA',
  team2: 'Argentina',
  team1Score: 118,
  team2Score: 97,
  venue: 'Chicago'
}
{
  id: ObjectId('6ac69002a44099e363056784')
}

```

## 17. SEARCH BY TEAM

MongoDB Query:

```
db.matches.find({
  team1:"India"
})
```

**Explanation:** Finds all basketball matches where India is Team 1.

```
Type "it" for more
> db.matches.find({
  team1:"India"
})
< {
  _id: ObjectId('6ac69002a44099e36305676e'),
  matchId: 1,
  team1: 'India',
  team2: 'Australia',
  team1Score: 92,
  team2Score: 87,
  venue: 'New Delhi'
}
{
  _id: ObjectId('6ac69002a44099e363056772'),
  matchId: 5,
  team1: 'India',
  team2: 'Japan',
  team1Score: 105,
  team2Score: 95,
  venue: 'Mumbai'
}
```

## 18. COUNT DOCUMENTS

**MongoDB Query:**

**db.matches.countDocuments()**

**Explanation:** Counts the documents currently present in the collection after the delete operations.

```
> db.matches.countDocuments()
```

```
< 38
```

```
BasketballDB>
```